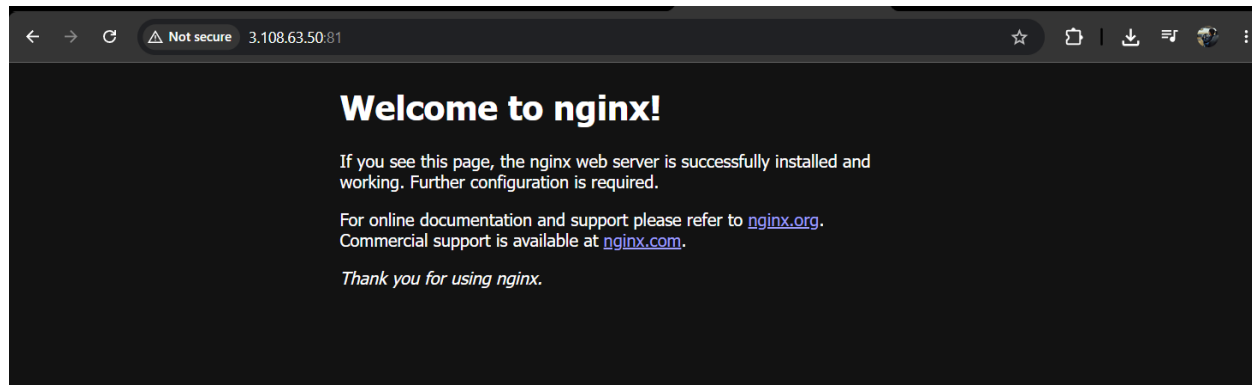


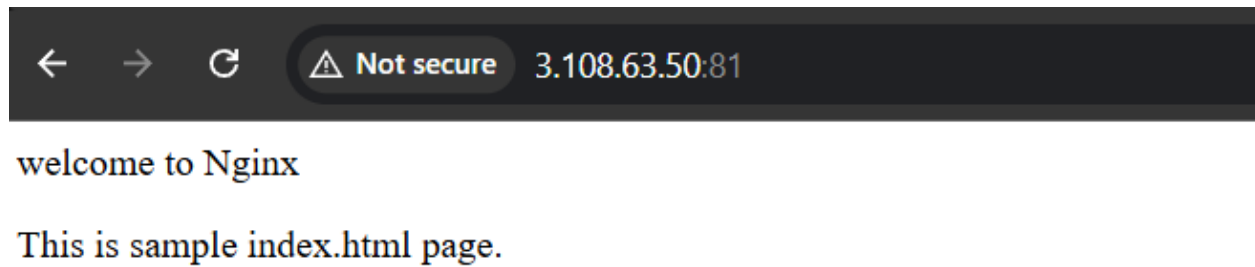
1. Install nginx and run nginx on port number 81.

```
root@ip-172-31-12-187:/home/ubuntu# lsof -i tcp:81
COMMAND  PID    USER  FD   TYPE    DEVICE  SIZE/OFF  NODE NAME
nginx    3365   root   5u    IPv4    27554      0t0     TCP *:81 (LISTEN)
nginx    3365   root   6u    IPv6    27555      0t0     TCP *:81 (LISTEN)
nginx    3366  www-data 5u    IPv4    27554      0t0     TCP *:81 (LISTEN)
nginx    3366  www-data 6u    IPv6    27555      0t0     TCP *:81 (LISTEN)
nginx    3367  www-data 5u    IPv4    27554      0t0     TCP *:81 (LISTEN)
nginx    3367  www-data 6u    IPv6    27555      0t0     TCP *:81 (LISTEN)
```

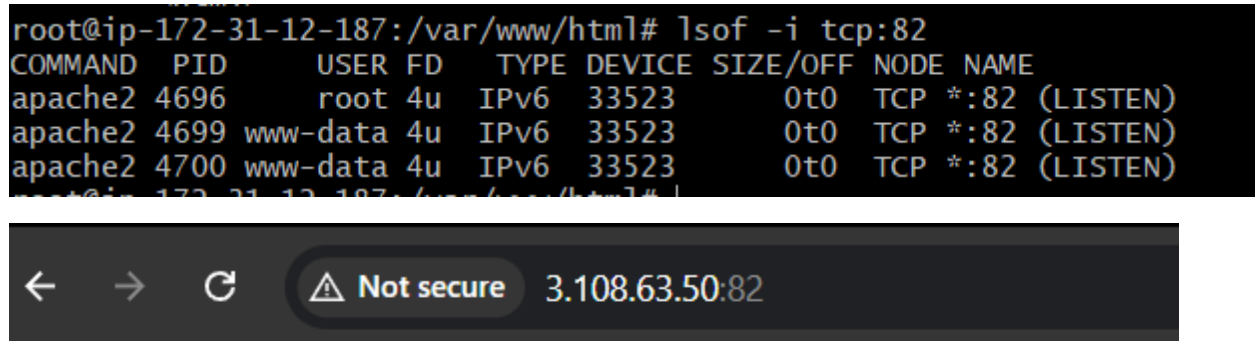


2. Deploy a sample index.html file on nginx.

```
root@ip-172-31-12-187:/home/ubuntu# cd /var/www/html
root@ip-172-31-12-187:/var/www/html# ls -l
total 4
-rw-r--r-- 1 root root 615 Aug 13 18:45 index.nginx-debian.html
root@ip-172-31-12-187:/var/www/html# sudo vi /var/www/html/index.html
root@ip-172-31-12-187:/var/www/html# cd /var/www/html
root@ip-172-31-12-187:/var/www/html# sudo vi index.html
root@ip-172-31-12-187:/var/www/html# cat /var/www/html/index.html
<html>
  <head>
    <title> This is Asifshaik's Nginx Website</title>
  </head>
  <body>
    <h1>welcome to Nginx</h1>
    <p>This is sample index.html page.</p>
  </body>
</html>
```



3. Install Apache and run Apache on port number 82



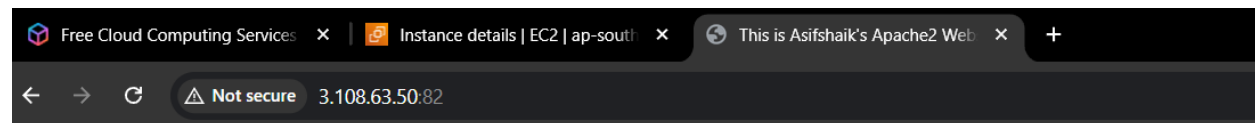
4. Deploy a sample index.html file on Apache.

```

ubuntu@ip-172-31-12-187:~$ sudo nano /var/www/html/index.html
ubuntu@ip-172-31-12-187:~$ cat /var/www/html/index.html
<html>

  <head>
    <title> This is Asifshaik's Apache2 Website</title>
  </head>
  <body>
    <h1>welcome to Apache2</h1>
    <p>This is sample index.html page created by Asifshaik.</p>
  </body>
</html>

```



welcome to Apache2

This is sample index.html page created by Asifshaik.

5. Install Apache tomcat on port number 8082

Home Documentation Configuration Examples Wiki Mailing Lists Find Help

Apache Tomcat/10.1.57

If you're seeing this, you've successfully installed Tomcat. Congratulations!

 Recommended Reading:

- [Security Considerations How-To](#)
- [Manager Application How-To](#)
- [Clustering/Session Replication How-To](#)

Server Status
Manager App
Host Manager

Developer Quick Start

- [Tomcat Setup](#)
- [First Web Application](#)
- [Realms & AAA](#)
- [JDBC DataSources](#)
- [Examples](#)
- [Servlet Specifications](#)
- [Tomcat Versions](#)

Managing Tomcat

For security, access to the [manager webapp](#) is restricted. Users are defined in:

```
$CATALINA_HOME/conf/tomcat-users.xml
```

In Tomcat 10.1 access to the manager application is split between different users. [Read more...](#)

[Release Notes](#)
[Changelog](#)
[Migration Guide](#)
[Security Notices](#)

Documentation

[Tomcat 10.1 Documentation](#)
[Tomcat 10.1 Configuration](#)
[Tomcat Wiki](#)

Find additional important configuration information in:

```
$CATALINA_HOME/RUNNING.txt
```

Developers may be interested in:

- [Tomcat 10.1 Bug Database](#)
- [Tomcat 10.1 JavaDocs](#)
- [Tomcat 10.1 Git Repository at GitHub](#)

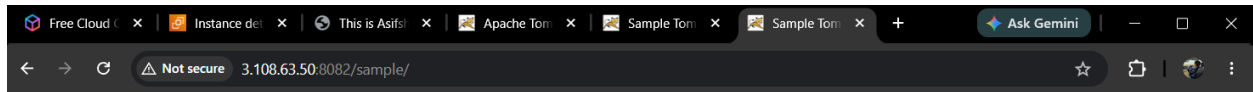
Getting Help

FAQ and Mailing Lists

The following mailing lists are available:

- [tomcat-announce](#)
Important announcements, releases, security vulnerability notifications. (Low volume).
- [tomcat-users](#)
User support and discussion
- [taglibs-user](#)
User support and discussion for [Apache Taglibs](#)
- [tomcat-dev](#)
Development mailing list, including commit messages

6. Deploy a sample app on webapps



Welcome to Asif shaik's Sample Application

Tomcat is running successfully on port 8082.

7. Create a tomcat.service file for tomcat.

```
ubuntu@ip-172-31-12-187:/opt$ sudo nano /etc/systemd/system/tomcat.service
ubuntu@ip-172-31-12-187:/opt$
```

```
ubuntu@ip-172-31-12-187:/opt
GNU nano 8.7.1 /etc
[Unit]
Description=Apache Tomcat Web Application Container
After=network.target

[Service]
Type=simple

Environment="JAVA_HOME=/usr/lib/jvm/java-17-openjdk-amd64"
Environment="CATALINA_HOME=/opt/tomcat"
Environment="CATALINA_BASE=/opt/tomcat"

ExecStart=/opt/tomcat/bin/catalina.sh run
ExecStop=/opt/tomcat/bin/shutdown.sh

Restart=on-failure

[Install]
WantedBy=multi-user.target
```